P1: Probabilistic electricity-price forecasting under regime shift

Corrected local edition v1.1.0 - 18 October 2026

Technical verification and supplementary control update - 21 October 2026

Publication snapshot - 23 October 2026; experiment results unchanged

SIMULATED BENCHMARK - NOT FIELD DATA. This report concerns a research-training portfolio, not a named electricity market, employer, Tarim Oilfield or trading system. This report documents the correction, automated tests and saved reference experiments. Earlier v1.0.2 archives remain in the development repository and are not included in this publication snapshot.

Purpose and experimental design

The question is how raw quantile forecasts, static conformal calibration and a rolling adaptive wrapper trade coverage against interval width when a simulated electricity-price process changes regime. This is an implementation and evaluation exercise using established methods; no new foundational algorithm is proposed.

The simulator produces 420 hourly days from 1 January 2024, with stable, volatile and renewable-shift regimes. Its 10,080 rows yield 9,888 direct forecast examples after eight days of history. Chronological issue-day splits contain 247 training, 82 calibration and 83 final-evaluation days. The final 1,992 targets span 3 December 2024-23 February 2025 and all belong to the renewable-shift regime. The final block is reused for gamma sensitivities and is not an untouched test set after that exploration.

Seven pooled histogram-gradient-boosting models estimate quantiles .05, .10, .25, .50, .75, .90 and .95, using horizon as a feature. Each has 140 iterations; row-wise sorting repairs quantile crossings. Features combine calendar variables, past price summaries and simulated forecast proxies. Future realised covariates and regime labels are excluded from model features. Static CQR uses either pooled calibration scores or separate horizon scores. The adaptive wrapper retains 30 scores per horizon, target miscoverage .20 and reference gamma .015.

Corrected baseline: seed 20260817

Method	Coverage	Mean width	Interval score
Raw quantiles	.556225	11.896091	34.007247
Static CQR, pooled	.924197	27.580914	32.176642
Static CQR, by horizon	.923695	28.378878	33.032735
Adaptive CQR	.806225	21.274739	30.727504

Width is in EUR/MWh. Lower interval score is better. The median model MAE is 6.299118 versus 11.268044 for the seasonal-naive forecast; mean pinball loss is 2.016259. The reported approximate CRPS, 4.501505, integrates only the available .05-.95 quantile grid and omits the tails; it is not exact full-distribution CRPS. All methods share the same raw quantiles, so these point/quantile metrics cannot distinguish interval wrappers. This sample does not establish general superiority.

Why v1.1.0 corrects the conformal calculation

For calibration score s_i = max(lower_i - y_i, y_i - upper_i) and n finite scores, set k = ceil((n+1)*(1-alpha)). For 1 <= k <= n, the correction is sorted score s_(k), indexed k-1 in code. If k > n, the augmented conformal score is positive infinity and the interval becomes unbounded. Alpha must be finite and strictly between zero and one. Empty, non-vector or nonfinite scores are rejected.

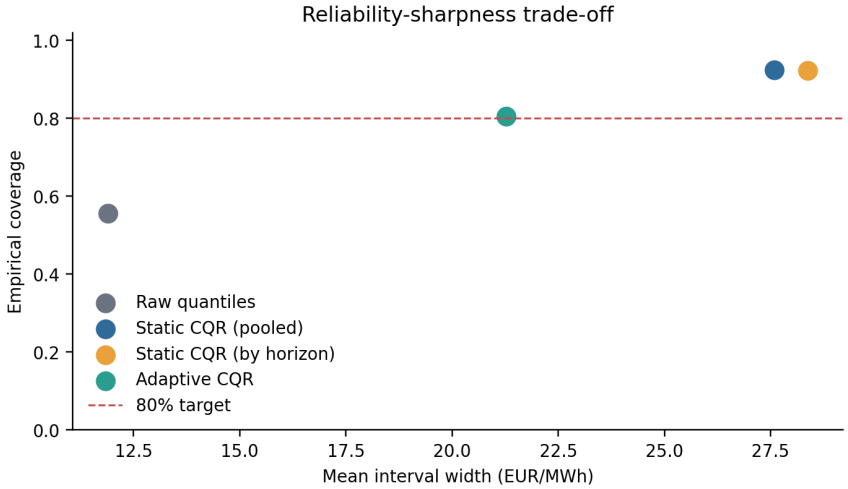
The old implementation passed k/n to np.quantile(..., method='higher'). NumPy maps that probability using n-1 positions, so this generally selects the next score. For scores 0-19, alpha .20 gives k=17: the correct value is 16, whereas the old code returns 17. It also capped k>n at the largest finite score, which cannot represent the required augmented infinity. Decimal alpha text is used for rank arithmetic to avoid .7 at n=9 becoming the wrong integer through binary roundoff; adjacent representable alpha values are tested separately.

The score convention is based on established conformalized quantile regression. Its standard finite-sample coverage justification requires the relevant exchangeability conditions. Changing the rank does not make those conditions hold in this dependent, shifted simulator.

Controlled comparison with preserved Windows v1.0.2

Method	Old coverage	New coverage	Old width	New width
Raw	.556225	.556225	11.896091	11.896091
Static pooled	.924197	.924197	27.600691	27.580914
Static horizon	.927711	.923695	29.282753	28.378878
Adaptive	.826305	.806225	22.362592	21.274739

The simulator fingerprint and all seven raw quantile columns match exactly across these two Windows runs. Adaptive interval score changes from 30.906253 to 30.727504. The original Python 3.12 reference is a separate historical run with raw coverage .547691; it is not mixed into this controlled comparison. The coverage-width chart now uses a full coverage axis and a regression test checks that the historical .547691 raw point is visible.



Corrected baseline coverage versus unconditional mean width.

Gamma sensitivity and seed robustness

The reference gamma remains .015. The .030 and .050 runs are exploratory sensitivities using the same seed, raw model and final block. They are not an independent basis for selecting a universally optimal adaptation rate.

Gamma	Adaptive coverage	Mean width	Interval score
.015	.806225	21.274739	30.727504
.030	.798193	21.107557	30.833048
.050	.795181	20.784492	30.999364

Here larger gamma slightly narrows intervals and reduces coverage; the interval score worsens. The corrected result is more cautious than a claim that faster adaptation necessarily improves reliability. Static and raw metrics are unchanged across these gamma runs, as expected.

Two additional simulation seeds use the entire unchanged baseline configuration, including 420 days, seven quantile models, gamma .015 and all three rolling folds. Together with the reference they give three baseline market simulations.

Seed	Raw coverage	Adaptive coverage	Adaptive width	Adaptive score
20260817	.556225	.806225	21.274739	30.727504
20260818	.539157	.814257	22.176603	32.236669
20260819	.558735	.817269	22.474140	32.780810

Adaptive coverage ranges .806225-.817269 across these three seeds. These runs are a limited simulator check, not a confidence interval for real-market coverage. Changing the seed also changes the model's random seed; this experiment does not separate model stochasticity from simulated-market variation.

Supplementary fixed-alpha rolling control - 21 October

The static by-horizon method fixes a correction using all 82 calibration days; the adaptive wrapper both rolls a 30-score buffer and updates alpha. A new gamma=0 control retains the identical rolling buffers, raw quantiles and alpha target .20, isolating feedback from the rolling-window effect. Each of the three original baseline models was reconstructed once, and its test raw quantiles and adaptive bounds matched the preserved reference exactly before comparison.

Seed	Fixed coverage	Fixed width	Fixed score	Adaptive score
20260817	.820281	21.768209	30.592811	30.727504
20260818	.822289	22.472269	32.070489	32.236669
20260819	.825803	22.681780	32.514927	32.780810

Feedback narrows widths by .208-.493 EUR/MWh and moves coverage nearer .80, but worsens interval score in all three seeds by .135-.266. These data do not establish an interval-score advantage for alpha feedback alone. The earlier adaptive/static comparison evaluates the combined wrapper. New shared raw frames, interval paths, metrics and actual execution records are in corrected_v110/supplementary_20261021/; this is not a new untouched test set or a rewrite of the earlier evidence.

Infinity and clipping are part of the method boundary

The update uses alpha_next = clip(alpha + gamma*(.20-error), .01, .99) and a rolling score distribution. It is inspired by Adaptive Conformal Inference, but clipping changes the unmodified recursion. The usual telescoping argument for that recursion cannot be transferred to this wrapper without a new analysis.

With 30 scores, an alpha below 1/31 requires an infinite conformal bound. The code honors that case, and the test suite verifies propagation into adaptive intervals. In all five completed runs, final-test unbounded fractions are zero. This observation does not justify replacing future infinities with finite caps. If an unbounded interval occurs, unconditional mean width and interval score remain infinite. Explicit finite_only_mean_width and finite_only_interval_score columns describe the finite subset only; they do not replace full-sample metrics.

Configuration and verification update

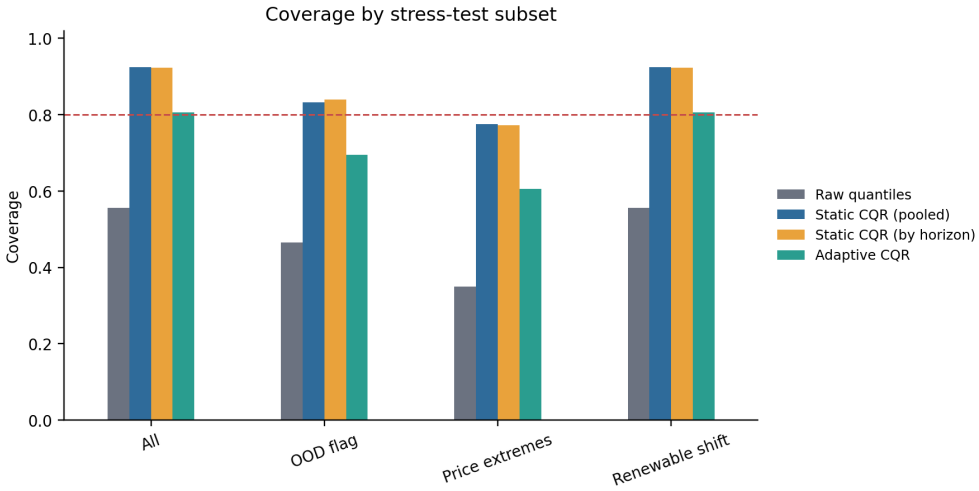
The 21 October suite passed 27 tests, adding a genuine small-model complete pipeline smoke at q05/q95 and a 90% target, configured plot/backtest checks, future-price/outcome perturbations, a fixed-alpha rolling control and protected output paths. A fresh complete default run reproduced all six baseline CSVs byte-for-byte, including the downstream P2 input. Separate read-only scripts check the five reference outputs and replay all three supplementary controls. The current suite has 28 tests, including the full-range rolling-coverage plot regression. The workflow runs on Ubuntu and Windows with Python 3.11 and 3.13. Results for this publication repository are recorded in its Actions tab and are separate from earlier development checks. These checks do not establish real-market validity or research novelty.

Stress subsets reveal conditional failures

Overall adaptive coverage close to .80 hides much poorer coverage among extremes and flagged examples. Price extremes are the lower and upper empirical deciles of the final outcomes, selected retrospectively for evaluation. They are not issuance-time labels. The OOD flag uses a training-standardised feature-distance score and a training 95th-percentile threshold; it is a simple diagnostic, not a validated detector of operational failure.

Subset	Rows	Raw coverage	Pooled coverage	Adaptive coverage
All	1,992	.556225	.924197	.806225
Price extremes	400	.350000	.775000	.605000
OOD flag	131	.465649	.832061	.694656

For price extremes, adaptive mean width is 22.455469 and score is 53.229701, compared with pooled width 29.173814 and score 47.900275. Adaptive calibration does not dominate pooled calibration on this subset. OOD adaptive width is 22.096972 and score 38.840207. These small, dependent subsets do not support precise inferential claims.



Coverage by stress subset. Renewable shift and All refer to the same final rows.

All final-evaluation rows are renewable-shift examples: Renewable shift repeats All. This table contains no volatile-regime rows and cannot establish performance in that regime. The rolling diagnostics below cover earlier periods.

Rolling-origin diagnostic

Each chronological expanding-window fold uses 42 calibration days, followed by 28 evaluation days (672 hours). Folds share one constructed timeline and are not independent replications.

Fold	Evaluation start	Raw coverage	Pooled coverage	Adaptive coverage
1	2024-07-16	.671131	.791667	.805060
2	2024-09-02	.486607	.578869	.720238
3	2024-10-20	.395833	.784226	.808036

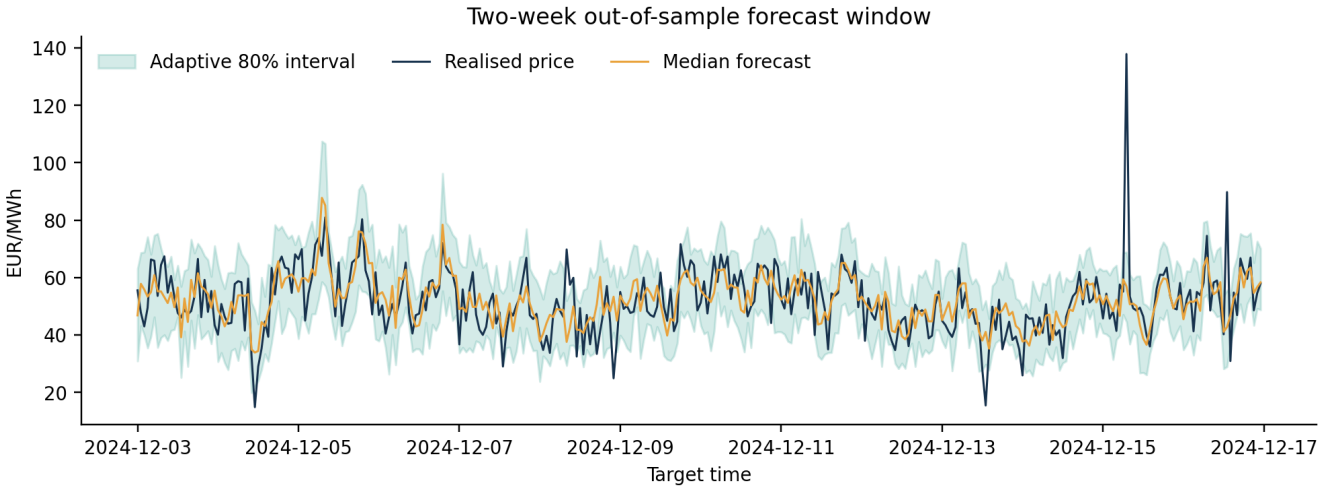
Fold 2 adaptive coverage is only .720238. Reporting the final .806225 alone would therefore hide a material reliability failure under a different segment of shift.

Timing and joint uncertainty limitations

The historical code calls the task day-ahead, but daily issue time is midnight and target horizon 1 is that same midnight. Thus horizon 1 is nowcast-like and the other targets have lead times 1-23 hours. A claim of operational 24-hour day-ahead forecasting requires an adapter with a genuine prior issuance cutoff.

Price lags and summary features use past history. Forward fill uses observations available before issuance; incomplete leading history is rejected rather than backfilled from future values. Nonetheless, the load, renewable and temperature forecast proxies are constructed from simulated realised series with added noise. They are assumed available at issuance and may be more informative than actual forecasts. There are no publication-vintage or revision records to prove real availability. This distinction must carry into any downstream scheduling study.

Adaptive buffers are separate by horizon. An earlier horizon's outcome on the same issue day does not update a later horizon's buffer. A horizon uses earlier days' outcomes before its next prediction, assuming immediate publication once the target is observed. Delays, settlement revisions and data outages require an explicit operational availability model before deployment.



First two weeks of corrected evaluation forecasts, with marginal adaptive intervals.

An 80% hourly marginal interval does not imply an 80% chance that all 24 hourly prices lie inside their intervals. These models do not calibrate a daily maximum score, a simultaneous band, a temporal copula or joint scenario distribution. Passing these marginal quantiles or widths into downstream optimisation does not establish joint price-path calibration, guarantee a dispatch risk level, or prove that a chosen CVaR objective is calibrated to realised losses.

Row-wise sorting also alters the original fitted quantile functions. Only seven quantile levels are available; tail shape between and beyond them is not learned by this report. The simple OOD score ignores temporal structure and feature correlations. These are concrete directions for validation, not claims of solved operational risk management.

Reproducibility, provenance and release status

Version metadata and runtime version are both 1.1.0. The original Windows Python 3.13.15 environment was used read-only; NumPy 2.3.5, pandas 2.2.3, SciPy 1.17.0 and scikit-learn 1.8.0 are recorded in manifests. Bytecode was disabled, plotting cache was redirected inside this workspace, and the corrected local src took priority over the original environment's editable installation. The reference experiments used a local source copy without writing to the Desktop or Downloads artifacts.

The original reference suite passed 16 tests, including exact discrete indices, alpha boundaries, ties, nonfinite/invalid inputs, infinity propagation, infinite scores and explicit conditional diagnostics. A live chart regression includes the old raw .547691 point. Existing checks cover chronological separation, forbidden future features, leading missing history, horizon separation, and an adaptive interval's independence from its own outcome. These tests check specific invariants; they are not a proof against every leakage route or every implementation defect.

Five full corrected experiment runs are saved under corrected_v110/: baseline, gamma .030, gamma .050, and additional baseline seeds 20260818/20260819. Each has a manifest, four tables, six figures, simulated data and forecast exports. Reference terminal logs and unit-test output are in corrected_v110/logs/. The dated manifests describe the source and artifacts used for those runs. Some entries refer to earlier source files or archives that are not distributed here; they are not an inventory of this publication snapshot. The separate PUBLICATION_MANIFEST.json lists the files in this snapshot. SHA256 identifies artifacts, not scientific validity. The Desktop working repository supplied only the read-only Python environment for the reference runs.

The baseline CSV has 1,992 rows and includes raw quantiles, pooled/horizon static bounds, adaptive bounds, alpha, correction and OOD columns. Its data fingerprint matches the original Windows run. The earlier v1.0.2 numerical outputs, report and reproduction records remain outside this snapshot in the development repository. This repository starts with the current release files, not the old Git history. Initial code and reference analyses were supplied; method sources and subsequent checks are described in PROJECT_HISTORY.md and RELATED_WORK.md. Documentation and report presentation wording was cleaned on 19 October 2026; the numerical artifacts were retained. The publication edit changes distribution notes, not the experiment dates, methods or numerical conclusions.

The corrected findings are narrower than a general performance claim: the rank defect is repaired and regression-tested; overall adaptive coverage is near .80 in the evaluated final blocks, while stress subsets and one rolling fold fail. Three simulated seeds do not establish real-market external validity, arbitrary time-series coverage, or jointly reliable daily paths. A next study would need real publication vintages, a true issuance gap, delayed-label handling, additional untouched temporal blocks and an explicitly validated joint uncertainty model.

Relevant verified primary references

Romano, Y., Patterson, E., and Candès, E. J. (2019). Conformalized Quantile Regression. Source of the quantile-based conformity score and static CQR framework.

Gibbs, I., and Candès, E. (2021). Adaptive Conformal Inference Under Distribution Shift. Source motivating the adaptive update; its guarantees are not asserted for this clipped, rolling implementation. Both primary-source records were checked on 18 October 2026. This report makes no priority claim beyond the local correction.